

From Sugar to Shelf Life: Exploring Cultured Dextrose as a Natural Preservative

Shubhkarman Singh Sodhi¹, Agnes Mukurumbira¹, Snehal Jadhav¹ and Nuwan Vithange²

¹ School of Exercise and Nutrition Sciences, Deakin University

² Senior Research Scientist, Noumi Limited

Email: s222627878@deakin.edu.au

Introduction:

Postharvest deterioration of citrus fruit contributes substantially to food loss and is associated with microbial spoilage, moisture loss, and deterioration of fruit quality(1). Citrus fruits are one of the most prominent food sources globally and make up a majority of food waste (2). Wax coatings are widely used to extend citrus shelf life by reducing moisture loss and modifying gas exchange; however, conventional wax coatings provide limited antimicrobial protection (3). Incorporating natural antimicrobial agents into wax coatings may provide an alternative approach for improving citrus preservation while reducing reliance on conventional chemical preservatives. Lemon myrtle essential oil (LMO)(4) and cultured dextrose (CD)(5) have demonstrated antimicrobial properties and may therefore be suitable candidates for incorporation into food-grade wax coating systems.

Methods:

This study was divided into three further sub studies which investigated the antimicrobial activity and postharvest preservation potential of food-grade wax coatings containing LMO and CD. In study one in vitro antimicrobial testing was conducted to evaluate the activity of both agents at various concentrations against relevant bacterial and fungal microorganisms. Based on the antimicrobial screening, study two used selected concentrations which were mixed with a shellac based wax and further tested for antimicrobial properties. Based on the second study, LMO formulation was applied to mandarin fruit alongside controls and evaluated during storage in study three. Microbial activity is being assessed at predetermined storage intervals, while changes in fruit quality are evaluated using measurements of weight loss, colour, firmness, and visible decay.

Results:

Study one: LMO shows complete inhibition of *Aspergillus Niger* and *Staphylococcus Aureus*. CD however did not show any inhibition.

Study two: Wax with CD did not show any antimicrobial properties, while LMO incorporated in Wax increased the susceptibility of the wax against pathogens at 4% of LMO. The susceptibility of LMO on its own is higher.

Study three: The experiment is still ongoing.

Conclusion: LMO has shown great antimicrobial resistance against common pathogens relevant to food spoilage. CD however has not shown any antimicrobial resistance at concentrations advised by the manufacturer. Its efficacy on mandarins would be concluded after study 3 has completed.

REFERENCES

1. Food Waste Index Report 2024 [Internet]. 2024 [cited 2026 Apr 19]. Available from: <https://wedocs.unep.org/handle/20.500.11822/45230>
2. Citrus | Food and Agriculture Organization of the United Nations [Internet]. [cited 2026 Apr 19]. Available from: [https://www.fao.org/markets-and-trade/commodities-overview/food-and-agriculture-market-analysis-\(FAMA\)/citrus/1/en](https://www.fao.org/markets-and-trade/commodities-overview/food-and-agriculture-market-analysis-(FAMA)/citrus/1/en)
3. Strano MC et al. Horticulturae 8: 612, 2022. doi:10.3390/horticulturae8070612
4. Rahman MM et al. J Hort Sci Biotechnol 97: 524-33, 2022. doi:10.1080/14620316.2021.2011433
5. Samapundo S et al. LWT Food Sci Technol 76: 101-7, 2017. doi:10.1016/j.lwt.2016.10.043